

## Appendix B: On the Age and Size of the Cosmos

This appendix establishes the mathematical constraints on the temporal age and spatial volume of the dual-manifold system under the topological assumption of a nested three-torus ( $T^3$ ). While the global geometry deviates from infinite flat space, the local presentation of the cosmological background matches observational constraints, ensuring consistency with the standard  $\Lambda$ CDM model parameters.

### B.2 The Cosmic Age Parameter ( $t_0$ )

To demonstrate why the local presentation of the cosmos yields a standard age of approximately 13.8 billion years, we evaluate the expansion rate within the standard Friedmann framework. Given that the local screening mechanism drives the effective coupling constant toward its general relativistic limit ( $g_{\text{eff}} \rightarrow 1$ ) in typical density regimes, the background Hubble parameter  $H(a)$  tracks the standard energy density evolution.

$$t_0 = \int_0^1 da / [a \cdot H(a)]$$

$$H(a) = H_0 \sqrt{[\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0}]}$$

$$t_0 = (2 / [3 H_0 \sqrt{\Omega_{\Lambda,0}}]) \cdot \ln([ \sqrt{\Omega_{\Lambda,0}} + 1 ] / \sqrt{\Omega_{m,0}})$$

$$t_0 \approx 13.787 \times 10^9 \text{ years}$$

### B.3 Spatial Scale and Volume of the Nested Torus ( $T^3 \times T^3$ )

The spatial framework is modeled as a nested topological manifold where a large, macroscopic three-torus contains or is influenced by a compactified, secondary shadow manifold.

#### Manifold A: The Large/Local Manifold ( $M_{\text{local}}$ )

The dimensions of our visible manifold are bounded below by the particle horizon ( $d_H$ ):

$$d_H = c \int_0^{t_0} dt / a(t) \approx 46.5 \text{ billion light-years}$$

$$L_{\text{local}} > 2 d_H \Rightarrow L_{\text{local}} \gtrsim 93 \times 10^9 \text{ light-years}$$

$$V_{local} = (L_{local})^3 \gtrsim 8.04 \times 10^{32} \text{ ly}^3$$